

ROTTNEST ISLAND, Australia

WMO: 946020

Lat: 32.0069S

Lon: 115.5022E

Elev: 43

StdP: 100.81

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	11.2	3.0	4.7	13.9	3.9	5.0	13.9	18.7	15.8	17.2	15.5	6.7	90	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.3	31.4	20.0	29.2	19.7	27.4	19.5	23.1	26.0	22.1	25.3	21.4	24.7	5.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.3	17.1	24.5	21.2	16.0	23.6	20.4	15.1	23.2	69.1	26.0	65.2	25.3	62.5	24.7	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	14.2	13.0	8.8	36.6	0.7	1.7	8.3	37.9	7.8	38.8	7.4	39.8	6.9	41.0
			6.4	23.9	0.8	1.8	5.8	25.2	5.3	26.3	4.8	27.3	4.2	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	22.2	22.8	22.0	20.2	18.1	16.2	15.1	15.2	15.7	17.0	19.2	20.8	(6)
(7)		DBStd	3.55	2.51	2.71	2.66	2.28	1.99	1.78	1.72	1.72	1.85	2.11	2.46	2.73	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	456	1	1	2	7	27	67	100	99	82	52	15	4	(9)
(10)		CDD10.0	3173	379	359	372	306	252	185	159	160	171	218	277	336	(10)
(11)		CDD18.3	586	121	126	115	63	21	2	1	1	2	11	42	82	(11)
(12)		CDH23.3	2183	487	619	461	124	16	0	0	0	3	30	128	314	(12)
(13)	CDH26.7	568	131	170	114	19	0	0	0	0	1	6	36	92	(13)	
(14)	Wind	WSAvg	7.5	8.3	8.0	7.5	6.6	6.5	7.1	7.3	7.0	7.2	7.5	8.0	8.5	(14)
(15)	Precipitation	PrecAvg	629	6	11	14	33	92	132	133	87	55	35	18	8	(15)
(16)		PrecMax	876	32	66	64	80	184	264	283	199	114	91	61	38	(16)
(17)		PrecMin	333	0	0	0	0	26	47	61	43	16	1	1	0	(17)
(18)		PrecStd	129	8	15	18	24	48	54	43	31	24	21	14	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	34.0	33.5	29.1	25.2	21.6	20.6	21.1	22.9	26.9	30.7	33.0	(19)
(20)			MCWB	20.2	21.0	20.7	18.3	17.0	15.5	15.9	16.0	15.5	16.7	17.8	19.3	(20)
(21)		2%	DB	29.9	31.0	29.5	26.3	23.3	20.5	19.4	19.7	20.6	23.3	26.7	29.2	(21)
(22)			MCWB	20.4	20.8	19.6	18.1	17.3	15.5	15.2	15.8	15.9	16.4	17.9	19.3	(22)
(23)		5%	DB	27.5	28.5	27.2	24.5	22.0	19.6	18.5	18.7	19.4	21.3	24.0	26.6	(23)
(24)			MCWB	20.1	20.5	19.2	18.3	17.0	15.2	14.6	15.0	15.5	16.7	17.5	19.0	(24)
(25)		10%	DB	25.6	26.6	25.6	23.2	21.1	18.9	17.7	17.9	18.5	20.1	22.5	24.5	(25)
(26)			MCWB	20.1	20.2	19.8	18.1	16.7	14.7	14.1	14.3	15.0	16.1	17.5	18.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.9	24.8	24.6	21.8	20.6	18.2	17.7	18.2	18.4	19.5	20.7	21.9	(27)
(28)			MCDWB	27.4	28.4	25.3	23.3	22.0	19.7	19.1	19.7	19.9	21.4	23.7	26.6	(28)
(29)		2%	WB	22.6	23.5	22.4	21.0	19.2	17.2	16.5	17.0	17.3	18.4	19.7	21.1	(29)
(30)			MCDWB	25.7	26.4	25.1	23.0	21.4	19.1	18.4	18.7	19.3	20.7	22.9	25.3	(30)
(31)		5%	WB	21.7	22.3	21.5	20.1	18.3	16.4	15.6	15.9	16.3	17.6	19.0	20.3	(31)
(32)			MCDWB	24.9	25.7	24.6	22.7	20.9	18.7	17.6	17.9	18.7	20.1	22.4	24.4	(32)
(33)		10%	WB	20.9	21.4	20.9	19.2	17.3	15.6	14.8	15.0	15.3	16.8	18.3	19.6	(33)
(34)			MCDWB	24.4	25.1	24.0	22.3	20.5	18.2	17.1	17.3	18.0	19.6	21.6	23.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.9	6.3	5.9	5.1	4.7	4.2	4.1	4.4	4.7	5.3	5.7	5.8	(35)
(36)			MCDBR	9.4	9.7	8.8	7.0	5.8	4.9	4.8	5.2	5.9	7.6	9.2	9.6	(36)
(37)			MCWBR	4.6	4.6	4.6	4.2	3.9	3.5	3.6	3.8	4.1	4.6	5.0	4.6	(37)
(38)		5% WB	MCDWB	7.2	7.5	6.2	5.6	4.9	4.4	4.3	4.5	5.4	6.1	7.1	7.5	(38)
(39)			MCWBR	4.4	4.2	4.3	4.0	4.2	4.2	4.0	4.2	4.4	4.2	4.5	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.339	0.325	0.318	0.309	0.303	0.302	0.309	0.326	0.326	0.327	0.328	(40)	
(41)		taud	2.605	2.622	2.659	2.661	2.658	2.652	2.652	2.627	2.556	2.575	2.594	2.611	(41)	
(42)		Ebn at Noon	1000	981	963	917	877	857	876	913	944	983	1005	1014	(42)	
(43)		Edh at Noon	103	98	89	79	71	67	70	81	96	102	104	103	(43)	
(44)	All-Sky Solar Radiation	RadAvg	8.20	7.39	6.10	4.44	3.30	2.67	2.83	3.72	5.00	6.52	7.77	8.40	(44)	
(45)		RadStd	0.35	0.24	0.29	0.26	0.21	0.19	0.18	0.19	0.30	0.24	0.30	0.30	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page